Indian Institute of Technology Ropar
Department of Mathematics

MA302 - Optimization Methods with Applications
Second Semester of the Academic Year 2025-26

Tutorial sheet - 5

---

1. Write the dual to the following LP problem.

(a) **Maximize**

$$Z = x_1 - x_2 + 3x_3$$

subject to

$$\begin{aligned} &x_1 + x_2 + x_3 \leq 10, \\ &2x_1 - x_2 - x_3 \leq 2, \\ &2x_1 - 2x_2 - 3x_3 \leq 6, \end{aligned}$$

$$x_1, x_2, x_3 \geq 0.$$

(b) **Minimize**

$$Z = 3x_1 - 2x_2 + 4x_3$$

subject to

$$\begin{aligned} &3x_1 + 5x_2 + 4x_3 \geq 7, \\ &6x_1 + x_2 + 3x_3 \geq 4, \\ &7x_1 - 2x_2 - x_3 \leq 10, \\ &x_1 - 2x_2 + 5x_3 \geq 3, \\ &4x_1 + 7x_2 - 2x_3 \geq 2, \end{aligned}$$

$$x_1, x_2, x_3 \geq 0.$$

(c) **Minimize**

$$Z = x_1 + 2x_2$$

subject to

$$\begin{aligned} &2x_1 + 4x_2 \leq 160, \\ &x_1 - x_2 = 30, \\ &x_1 \geq 10, \end{aligned}$$

$$x_1, x_2 \geq 0.$$

(d) **Minimize**

$$Z = x_1 - 3x_2 - 2x_3$$

subject to

$$\begin{aligned}
&3x_1 - x_2 + 2x_3 \le 7,\\
&2x_1 - 4x_2 \ge 12,\\
&-4x_1 + 3x_2 + 8x_3 = 10,
\end{aligned}$$

$$x_1, x_2 \ge 0, \qquad x_3 \text{ unrestricted in sign.}$$

2. Write the dual of the following LPP (primal) and find the optimal solution of the dual by solving the primal LPP.

$$\textbf{Maximize } Z = 4x_1 + 3x_2$$

subject to

$$\begin{aligned}
x_1 + x_2 &\le 8,\\
2x_1 + x_2 &\le 10,\\
x_1, x_2 &\ge 0.
\end{aligned}$$

3. Write the dual of the following LPP (primal) and determine its optimal solution by solving the primal problem.

$$\textbf{Maximize } Z = -2x_1 - x_2$$

subject to

$$\begin{aligned}
&3x_1 + x_2 = 3,\\
&4x_1 + 3x_2 \ge 6,\\
&x_1 + 2x_2 \le 3,\\
&x_1, x_2 \ge 0.
\end{aligned}$$